



JABchem



Not to be shared without the copyright holder's permission

Past Papers Nat 5 Chemistry

2018 Marking Scheme

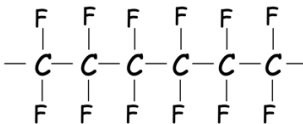
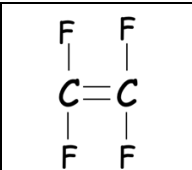
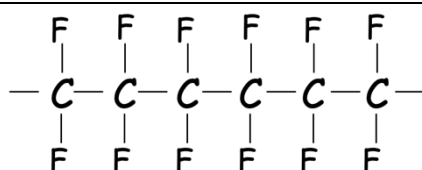
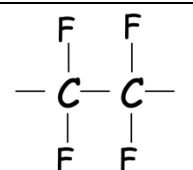
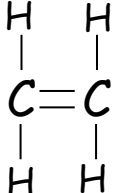
Grade Awarded	Mark Required		% candidates achieving grade
	(/125)	%	
A	92+	73.6%	36.3%
B	78+	62.4%	21.8%
C	64+	51.2%	19.1%
D	50+	40%	14.0%
No award	<50	<40%	8.8%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	17.6 /25	45.6 /75	18.2 /25

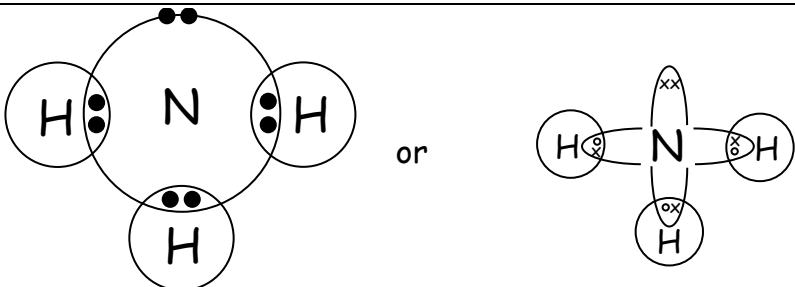
2018 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning																
1	A	91	☒ A Increasing particle size decreases the reaction rate ☒ B Increasing particle size increases the reaction rate ☒ C Increasing concentration increases the reaction rate ☒ D Adding a catalyst increases the reaction rate																
2	B	95	<table border="1"> <thead> <tr> <th>Particle</th><th>Location</th><th>Charge</th><th>Mass</th></tr> </thead> <tbody> <tr> <td>Proton</td><td>in nucleus</td><td>+1</td><td>1 amu</td></tr> <tr> <td>Neutron</td><td>in Nucleus</td><td>0</td><td>1 amu</td></tr> <tr> <td>Electron</td><td>outside nucleus</td><td>-1</td><td>Approx 0</td></tr> </tbody> </table>	Particle	Location	Charge	Mass	Proton	in nucleus	+1	1 amu	Neutron	in Nucleus	0	1 amu	Electron	outside nucleus	-1	Approx 0
Particle	Location	Charge	Mass																
Proton	in nucleus	+1	1 amu																
Neutron	in Nucleus	0	1 amu																
Electron	outside nucleus	-1	Approx 0																
3	B	82	☒ A Oxygen exists as diatomic O_2 molecules ☒ B Helium is a monatomic noble gas in group 0 ☒ C Bromine exists as diatomic Br_2 molecules ☒ D Hydrogen exists as diatomic H_2 molecules																
4	D	57	Hydrogen bromide has the formula HBr using the cross-over rule. The correct structure for this can only be linear.																
5	A	52	☒ A Na atoms have electron arrangement 2,8,1 $\therefore$ Na^+ ions have arrangement 2,8 ☒ B Mg atoms have electron arrangement 2,8,2 $\therefore$ Mg^+ ions have arrangement 2,8,1 ☒ C F atoms have electron arrangement 2,7 $\therefore$ F^+ ions have arrangement 2,6 ☒ D Ne atoms have electron arrangement 2,8 $\therefore$ Na^+ ions have arrangement 2,7																
6	D	67	☒ A Bonding type is covalent molecular as it does not conduct and has low mpt/bpt ☒ B Bonding type is ionic as it does not conduct as solid but does conduct as liquid ☒ C Bonding type is metallic as it conducts both as a solid and a liquid ☒ D Bonding type is covalent network as it does not conduct and has very high mpt																
7	C	57	$\text{concentration} = \frac{\text{no. of moles}}{\text{volume}} = \frac{0.1 \text{ mol}}{0.25 \text{ litres}} = 0.4 \text{ mol l}^{-1}$																
8	B	82	☒ A An alkaline solution contains more OH^- ions than H^+ ions (still contains some H^+) ☒ B An alkaline solution contains more OH^- ions than H^+ ions ☒ C An acidic solution contains more H^+ ions than OH^- ions ☒ D A neutral solution contains equal numbers of H^+ ions and OH^- ions																
9	C	64	☒ A Diluting acids with water increases pH until it reaches pH=7 ☒ B Diluting acids with water increases pH until it reaches pH=7 ☒ C Diluting acids with water decreases the H^+ ion concentration as water is added ☒ D Diluting acids with water decreases the H^+ ion concentration as water is added																
10	D	84	C_9H_{20} structure drawn is the isomer 3,4-dimethylheptane ☒ A C_9H_{20} structure is 2,4-dimethylheptane $\therefore$ different isomer not same structure ☒ B C_8H_{18} structure is 3,4-dimethylhexane $\therefore$ different formula not same structure ☒ C C_9H_{20} structure is 3,3-dimethylheptane $\therefore$ different isomer not same structure ☒ D C_9H_{20} structure is 3,4-dimethylheptane $\therefore$ same formula and same structure																
11	C	75	C_6H_{14} structure drawn is 2-methylpentane ☒ A Cyclohexane C_6H_{12} has different formula so cannot be an isomer of C_6H_{14} . ☒ B 2-methylpentane again but drawn different so cannot be an isomer. ☒ C 3-methylpentane C_6H_{14} so same formula but different structure so is an isomer. ☒ D 2-methylbutane C_5H_{12} has different formula so cannot be an isomer of C_6H_{14} .																
12	C	46	☒ A Hydrogenation: Adding hydrogen across a $C=C$ double bond to form alkane ☒ B Combustion: burning compound in oxygen to form CO_2 and H_2O ☒ C Hydration: Adding H_2O across $C=C$ double bond to form alcohol ☒ D Reduction: Gaining electrons																

23	D	70	☐ A covalent bonding contains a shared pair of electrons and two nuclei ☐ B there is no attraction between negative ions and electrons (they repel) ☐ C ionic bonding is the force of attraction between negative ions and positive ions ☒ D metallic bonding is the force of attraction between positive ions and delocalised electrons
24	B	53	☐ A neutralisation: reaction of H^+ ions to form H_2O ☒ B precipitation: two ions combining to form an insoluble solid. ☐ C addition: adding a molecule across a $C=C$ double bond ☐ D redox: electrons are transferred between reduction and oxidation reactions
25	D	69	☐ A Filtration (Step Z) must occur before evaporation (Step X) ☐ B Neutralisation Step Y must be first step ☐ C Neutralisation Step Y must be first step ☒ D Order: Neutralisation (Y) followed by Filtration (Z) followed by Evaporation (X)

2018 National 5 Chemistry Marking Scheme				
Long Qu	Answer	Reasoning		
1a	Carbon dioxide	hydrochloric acid + calcium carbonate → calcium chloride + water + carbon dioxide ACID + METAL CARBONATE → SALT + WATER + CARBON DIOXIDE		
1b(i)	0.5 cm ³ s ⁻¹	Rate = $\frac{\Delta \text{Quantity}}{\Delta \text{Time}} = \frac{77 - 62 \text{ cm}^3}{50 - 20 \text{ s}} = 0.5 \text{ cm}^3 \text{ s}^{-1}$		
1b(ii)	Graph showing:	1 mark	1 mark	1 mark
		One mark is awarded for a graph which shows points plotted rather than bars.	The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales	The axes of the graph have suitable labels and units.
1b(iii)	68±1	Problem Solving: Reading information from a graph.		
1c	One answer from:	Greater number of hydrogen ions/H ⁺	Greater concentration of hydrogen ions/H ⁺	Greater moles of hydrogen ions/H ⁺
2a(i)	addition	Addition reactions involve the opening up of the 2 nd bond in a C=C double bond and single bonds being formed on either side. Addition Polymerisation is when the C=C double bond in the monomer opens up and joins with other monomers to form a long polymer chain.		
2a(ii)		 Monomer	 Polymer	 Repeating Unit
2b		A addition monomer always has the shape: The monomer given in the question has four fluorine atoms in its corners. The other monomer has four hydrogen atoms in its corners.		
3a	Increase the percentage of carbon increases the average heat content	Problem Solving: Spotting a relationship in a table		
3b	46.7	gfm FeS ₂ : (1×56) + (2×32) = 56 + 64 = 120 (1 mark) % Fe = $\frac{56}{120} \times 100 = 46.7\%$ (1 mark)		
4a	carbon & hydrogen	A hydrocarbon is a compound containing the elements carbon and hydrogen only.		
4b	2-methylpropane or methylpropane	The longest continuous chain of carbons is 3 ∴ name ends in <i>propane</i> A 1 carbon side group is called a <i>methyl</i> ∴ <i>methylpropane</i> The methyl group is on 2 nd carbon ∴ 2-methylpropane (It is acceptable to drop the number 2 as the methyl group can only be positioned on carbon number 2)		

4c	stronger intermolecular	The process of evaporation/boiling does not change the strong covalent bonds inside a molecule. Boiling/evaporation is dependent on the weaker intermolecular bonds between molecules. The higher boiling point of butane compared to isobutene is due to the stronger intermolecular bonds between butane molecules compared to isobutene molecules.																															
4d	150-154°C	<table><tr><td>Alkane</td><td>Pentane</td><td>Hexane</td><td>Heptane</td><td>Octane</td><td>Nonane</td></tr><tr><td>Boiling Point (°C)</td><td>36</td><td>69</td><td>98</td><td>126</td><td>-</td></tr><tr><td>Difference:</td><td></td><td>33</td><td>29</td><td>28</td><td>Prediction: 27</td></tr><tr><td>Prediction:</td><td>-</td><td>-</td><td>-</td><td>-</td><td>153</td></tr></table>	Alkane	Pentane	Hexane	Heptane	Octane	Nonane	Boiling Point (°C)	36	69	98	126	-	Difference:		33	29	28	Prediction: 27	Prediction:	-	-	-	-	153							
Alkane	Pentane	Hexane	Heptane	Octane	Nonane																												
Boiling Point (°C)	36	69	98	126	-																												
Difference:		33	29	28	Prediction: 27																												
Prediction:	-	-	-	-	153																												
5a	Sodium azide, potassium nitrate and silicon dioxide	Problem Solving: Gathering information from a passage																															
5b	Potassium	<table><tr><td>Element</td><td>Ion</td><td>Flame Colour</td></tr><tr><td>Barium</td><td>Ba²⁺</td><td>Green</td></tr><tr><td>Potassium</td><td>K⁺</td><td>Lilac</td></tr><tr><td>Calcium</td><td>Ca²⁺</td><td>Orange-red</td></tr><tr><td>Sodium</td><td>Na⁺</td><td>Yellow</td></tr></table>	Element	Ion	Flame Colour	Barium	Ba ²⁺	Green	Potassium	K ⁺	Lilac	Calcium	Ca ²⁺	Orange-red	Sodium	Na ⁺	Yellow	<table><tr><td>Element</td><td>Ion</td><td>Flame Colour</td></tr><tr><td>Copper</td><td>Cu²⁺</td><td>Blue-green</td></tr><tr><td>Strontium</td><td>Sr²⁺</td><td>Red</td></tr><tr><td>Lithium</td><td>Li⁺</td><td>Red</td></tr></table>	Element	Ion	Flame Colour	Copper	Cu ²⁺	Blue-green	Strontium	Sr ²⁺	Red	Lithium	Li ⁺	Red			
Element	Ion	Flame Colour																															
Barium	Ba ²⁺	Green																															
Potassium	K ⁺	Lilac																															
Calcium	Ca ²⁺	Orange-red																															
Sodium	Na ⁺	Yellow																															
Element	Ion	Flame Colour																															
Copper	Cu ²⁺	Blue-green																															
Strontium	Sr ²⁺	Red																															
Lithium	Li ⁺	Red																															
5c	SiO ₂	<table><tr><td>Write down Valency below each element's symbol</td><td>Put in Cross-over Arrows</td><td>Follow arrows to get formula</td></tr><tr><td> Si O 4 2 </td><td> Si O 4 2 </td><td> Si₂O₄ Cancel Down SiO₂ </td></tr></table>	Write down Valency below each element's symbol	Put in Cross-over Arrows	Follow arrows to get formula	Si O 4 2	Si O 4 2	Si₂O₄ Cancel Down SiO₂																									
Write down Valency below each element's symbol	Put in Cross-over Arrows	Follow arrows to get formula																															
Si O 4 2	Si O 4 2	Si₂O₄ Cancel Down SiO₂																															
5d	44	Total volume given off is the maximum height attained by the graph.																															
6a(i)	2	The two lines on the graph represent the two different isotopes of boron with mass numbers of 10 and 11.																															
6a(ii)	10.8	$\text{ram} = \frac{(10 \times 20) + (11 \times 80)}{100} = \frac{200 + 880}{100} = 10.8$																															
6b	¹⁴ ₆ C	Atomic number = number of protons = 6 Mass number = no. of protons + no. of neutrons = 6+8 = 14																															
7a	Lattice	Ionic compounds have a structure of alternating positive and negative ions in each direction. This structure is called a lattice structure.																															
7b(i)	Ions are free to move	In the solid state, ions are locked together in a lattice structure and cannot move. The circuit cannot be completed without the movement of ions. In the liquid/molten state or the solution state, ions are free to move and will complete the circuit.																															
7b(ii)	oxidation	<table><tr><td>2Cl⁻(aq)</td><td>→</td><td>Cl₂(g)</td><td>+</td><td>2e⁻</td></tr><tr><td>chloride ions</td><td></td><td>chlorine gas</td><td></td><td>Loss of two electrons by the reactants</td></tr></table>					2Cl ⁻ (aq)	→	Cl ₂ (g)	+	2e ⁻	chloride ions		chlorine gas		Loss of two electrons by the reactants																	
2Cl ⁻ (aq)	→	Cl ₂ (g)	+	2e ⁻																													
chloride ions		chlorine gas		Loss of two electrons by the reactants																													
7b(iii)	One answer from:	Allows products to be identified	To make sure that only one product is produced at each electrode	To separate the strontium from the chlorine																													
8	Open Question:	<table><tr><td>3 mark answer</td><td>2 mark answer</td><td>1 mark answer</td></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the																									
3 mark answer	2 mark answer	1 mark answer																															
Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the																															

		involved and the application of these to respond to the problem.		chemistry within the problem is understood.
9a(i)	contains C=C double bond	Saturated: All bonds between carbons are single bonds and it does not decolourise bromine solution quickly. Unsaturated: At least one C=C double bond between carbons and will decolourise bromine solution quickly.		
9a(ii)	Bromine solution decolourises	Bromine Br ₂ adds across a C=C double bond by addition reaction. Each bromine joins across where the C=C double bond used to be and leaves a C-C single bond in its place. Bromine solution is yellow/orange and decolourises (loses its colour) when it adds across the C=C double bond.		
9b	472.8	heat energy = specific heat capacity × mass × change in Temperature $E_h = c \times m \times \Delta T$ $E_h = 1.97 \text{ kJ kg}^{-1} \text{ } ^\circ\text{C}^{-1} \times 1.5\text{kg} \times 20^\circ\text{C}$ $E_h = 472.8 \text{ kJ}$		
10a(i)	$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$	$\text{N}_2 + \text{H}_2 \rightleftharpoons \text{NH}_3$ 2xN atoms on LHS but only 1xN atom on RHS ∴ Double NH ₃ $\text{N}_2 + \text{H}_2 \rightleftharpoons 2\text{NH}_3$ 2xH atoms on LHS but 6xH atoms on RHS ∴ treble H ₂ $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$		
10a(ii)	One answer from:	Reaction is reversible	Reaction occurs in both directions	The reaction is at equilibrium
10b	Answer showing:			
10c	Ostwald Process	ammonia + oxygen $\xrightarrow{\text{platinum catalyst}}$ nitrogen monoxide + water $4\text{NH}_3 + 5\text{O}_2 \longrightarrow 4\text{NO} + 6\text{H}_2\text{O}$ $\downarrow \text{oxygen}$ NO_2 $\downarrow \text{water}$ Nitric Acid		
10d	ammonium nitrate	ammonia + water $\longrightarrow$ ammonium hydroxide ammonium hydroxide + nitric acid $\longrightarrow$ ammonium nitrate + water		
11a	Arrow showing flow through wires from magnesium/right to copper/left	Electrons travel through wires while ions flow through the solution. Electrons travel from higher metal in electrochemical series (magnesium on right) to metal lower down electrochemical series (copper on left).		

11b	One answer from:	Completes the circuit/cell	allows ions to flow/move/transfer (between the two beakers)	provide ions to complete circuit/cell		
11c	$\text{Cu}^{2+} + 2\text{e}^{-} \longrightarrow \text{Cu}$	Reduction is the gain of electrons by the reactant. Cu^{2+} ions will gain 2 electrons to become Cu metal.				
11d	Insoluble	Magnesium phosphate is insoluble (p8 of data booklet) and the ions are not able to move so the circuit would not be complete.				
12a	Alpha Particles are stopped by paper	Radiation	Alpha	Beta	Gamma	
		Mass	4	0	No mass	
		Charge	2	-1	No charge	
		Stopped by	Paper	Aluminium	Thick lead	
		Deflection	Towards negative	Towards positive	No deflection	
		Use	smoke detectors	Measuring thickness of paper in paper mill	Radiotherapy cancer treatment	
12b	14.8	Number of half-lives		Fraction		
		0		1		
		1		$\frac{1}{2}$		
		2		$\frac{1}{4}$		
		3		$\frac{1}{8}$		
		4		$\frac{1}{16}$		
1 half-life = 3.7 years 4 half-lives = $4 \times 3.7 \text{ years} = 14.8 \text{ years}$						
12c	increases stays the same	${}^{204}_{81}\text{Tl} \rightarrow {}^{204}_{82}\text{Pb} + {}^0_{-1}\text{e}$				
13a(i)	Carboxyl group	$-\text{O}-\text{H}$ hydroxyl group		$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$ carboxyl group		
13a(ii)	134	Gfm $\text{C}_4\text{H}_6\text{O}_5 = (4 \times 12) + (6 \times 1) + (5 \times 16) = 48 + 6 + 80 = 134\text{g}$				
13b	As halogen atom goes down group 7 the acidity decreases.	Any correct statement linking acidity to the position of the halogen				
		The acidity (of the carboxylic acids) decreases going down the group		As you go (up) from iodine to fluorine the acidity increases	The one at the top (of the group) has the highest acidity	The one that has the lowest acidity is at the bottom (of the group)
13c	$\begin{array}{ccccccc} & \text{OH} & \text{H} & \text{H} & \text{H} & \text{H} & \\ & & & & & & \\ \text{O}=\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{H} & \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & \text{H}-\text{C}-\text{H} & & & \\ & & & & & & \\ & & & \text{H} & & & \end{array}$	The carbon with the hydroxyl -OH group changes. 2 hydrogens are removed and replaced with a C=O group forming a carboxyl -COOH group on that carbon. Any structure of 4-methylpentanoic acid is correct.				
14a	C	Sample C is only sample with chloride ion concentration below 0.25 g l^{-1}				
14b(i)	Pipette	A pipette is the most appropriate piece of equipment to measure exact volume of liquids. They are more accurate than using measuring cylinders and beakers.				
14b(ii)	One answer from:	2 & 4 are concordant	They are within 0.2 cm^3	They are within 0.1 or 0.1 apart	Titration 1 and 3 or the other two are not concordant or not within 0.2 of each other	

14b(iii)	0.000161	no. of moles = volume x concentration = 0.00805litres x 0.02mol l ⁻¹ = 0.000161mol							
15a	Diamond(s) anvil cell	Problem Solving Question							
15b(i)	1.9million or 1900000	1000gigapascals <====> 10million atmospheres 190 gigapascals <====> 10million atmospheres x ¹⁹⁰ /1000 = 1.9 million atmospheres							
15b(ii)	Light bulb would not light up	Pressurised sodium becomes an insulator. Circuit will not be complete and the bulb will not light up.							
15c	Fe ₂ O ₃ ↓ O ₂ + Fe ₅ O ₇	Problem Solving Question							
16a(i)	Homologous Series	Homologous Series are families of compounds that have similar chemical properties and a general formula.							
16a(ii)	C _n H _{2n+1} SH or C _n H _{2n+2} S or C _n H _{2n+1} HS	Name	Methanethiol	Ethanethiol	Propanethiol				
		Structure							
		Formula	CH ₃ SH	C ₂ H ₅ SH	C ₃ H ₇ SH				
		Relationship	If n=1, 2n+1=3	If n=2, 2n+1=5	If n=3, 2n+1=7				
		General Formula	C _n H _{2n+1} SH	C _n H _{2n+1} SH	C _n H _{2n+1} SH				
16b	Any answer from:	<table><tr><td>Sulphur oxide</td><td>Sulphur monoxide</td><td>Sulphur dioxide</td><td>Sulphur trioxide</td></tr></table>				Sulphur oxide	Sulphur monoxide	Sulphur dioxide	Sulphur trioxide
Sulphur oxide	Sulphur monoxide	Sulphur dioxide	Sulphur trioxide						
16c	960	gfm CH₃OH = (1x12)+(4x1)+(1x16) = 12+4+16 = 32g no. of mol = $\frac{\text{mass}}{\text{gfm}}$ = $\frac{640\text{g}}{32\text{g mol}^{-1}}$ = 20mol CH₃OH + H₂S —————> CH₃SH + H₂O 1mol							

2018 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	A	91	2	5	2	0	0
2	B	1	95	2	1	0	0
3	B	2	82	9	7	0	0
4	D	9	9	25	57	0	0
5	A	52	1	8	39	0	0
6	D	11	15	6	67	0	0
7	C	17	18	57	7	1	0
8	B	4	82	11	2	0	0
9	C	8	11	64	17	0	0
10	D	4	10	2	84	0	0
11	C	5	17	75	2	0	0
12	C	26	17	46	11	1	0
13	A	44	5	35	16	0	0
14	B	7	70	4	18	1	0
15	B	40	55	2	2	0	0
16	C	9	18	68	4	1	0
17	A	88	5	5	1	0	0
18	B	16	58	17	8	1	0
19	A	93	3	2	2	0	0
20	C	6	5	82	6	1	0
21	C	3	13	82	2	0	0
22	D	5	7	20	67	0	0
23	D	9	9	11	70	0	0
24	B	14	53	20	13	0	0
25	D	18	2	9	69	0	0